B.Tech Degree IV Semester Examination April 2012

ME 405 HYDRAULIC MACHINERY

(2006 Scheme)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer ***ALL*** questions)

$(8 \times 5 = 40)$

I. (a) What is Buckingham's Pi theorem?
 (b) Explain principles of Similitude.
 (c) List the factors to be considered for the selection of type and speed of turbines for a particular application.
 (d) Differentiate between reaction turbines and impulse turbines.
 (e) List the advantages of centrifugal pump over reciprocating pump.
 (f) Explain the advantages of using air vessel in reciprocating pump.
 (g) Sketch and explain a torque converter.
 (h) What is a surge tank? Explain its use.

PART B

$(4 \times 15 = 60)$

II. A jet of water of diameter 6cm, strikes a curved vane at its centre. The curved vane is moving with a velocity of 10 m/s in the direction of jet. If the velocity of the jet is 22m/s and it is deflected through an angle of 160°, determine: (i) force exerted on the vane in the direction of jet. (ii) power of jet (iii) efficiency of the jet.

OR

III. Stating necessary assumptions, derive the momentum equation.

IV. A reaction turbine has diameter of 1m and flow area 0.35m^2 at the inlet. It operates under a head of 60m. The speed of the runner is 400rpm. Angles made by the absolute and relative velocities at the inlet are 18° and 60° respectively with the tangential velocity of the wheel. Assuming radial discharge, find (i) discharge (ii) power developed (iii) hydraulic efficiency.

OR

V. Write notes on:
 (i) Speed regulation of turbines
 (ii) Draft tube
 (iii) Cavitation

VI. (a) Derive an expression for discharge, Q for a reciprocating pump.
 (b) Draw and explain an indicator diagram for a reciprocating pump. Also show the effect of acceleration and friction on it.

OR

VII. A centrifugal pump delivers 50 litres of water per second to a height of 15m through a 20m long pipe. Diameter of the pipe is 14cm. Overall efficiency is 72% and the coefficient of friction is 0.015. Determine the power needed to drive the pump.

VIII. Draw a neat sketch of
 (i) Hydraulic intensifier
 (ii) Hydraulic accumulator
and explain the difference between them.

OR

IX. Write notes on:
 (i) Gear pump
 (ii) Hydraulic ram

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